



DC – AI – Routing – Switching – Security – Wireless – Automation

Using Apstra Drain Mode

An HPE Juniper Networking Techpost

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HPE



Contents

1	Using Apstra Drain Mode	2
1.1	Introduction	2
1.2	Activating Drain Mode	3
1.3	Disabling Drain Mode	3
1.4	IBA Monitoring of Devices in Drain Mode	3
1.5	Example	3
1.5.1	Recommended Usage	4
1.6	Configuration Examples	4
1.6.1	Spine (L2 and L3 Blueprints)	4
1.6.2	Leaf (L2 Server Facing Ports with MLAG)	5
1.6.3	Leaf (L2 Server Facing Ports no MLAG)	8
1.6.4	Leaf (L3 Connected Servers)	11
1.7	Summary	11
1.8	Useful links	12
1.9	Glossary	12
1.10	Acknowledgements	12

1 Using Apstra Drain Mode

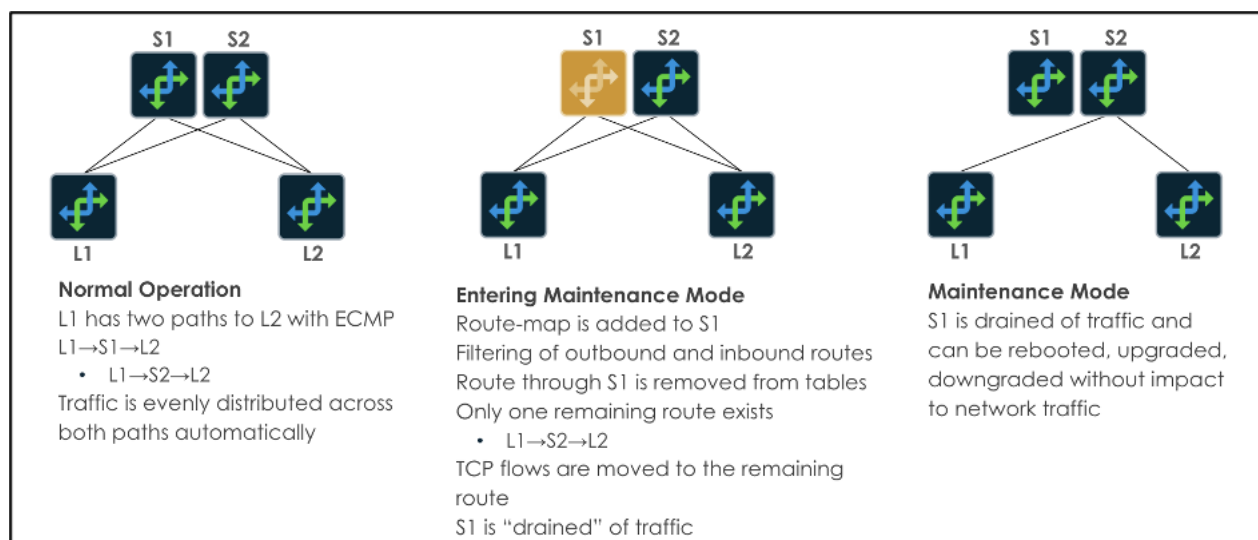
Jeffrey Doyle - 11/16/2023

Apstra supports Drain Mode for managed switches, allowing the operator to gracefully drain traffic from devices without simply shutting down the BGP neighbor relationships.

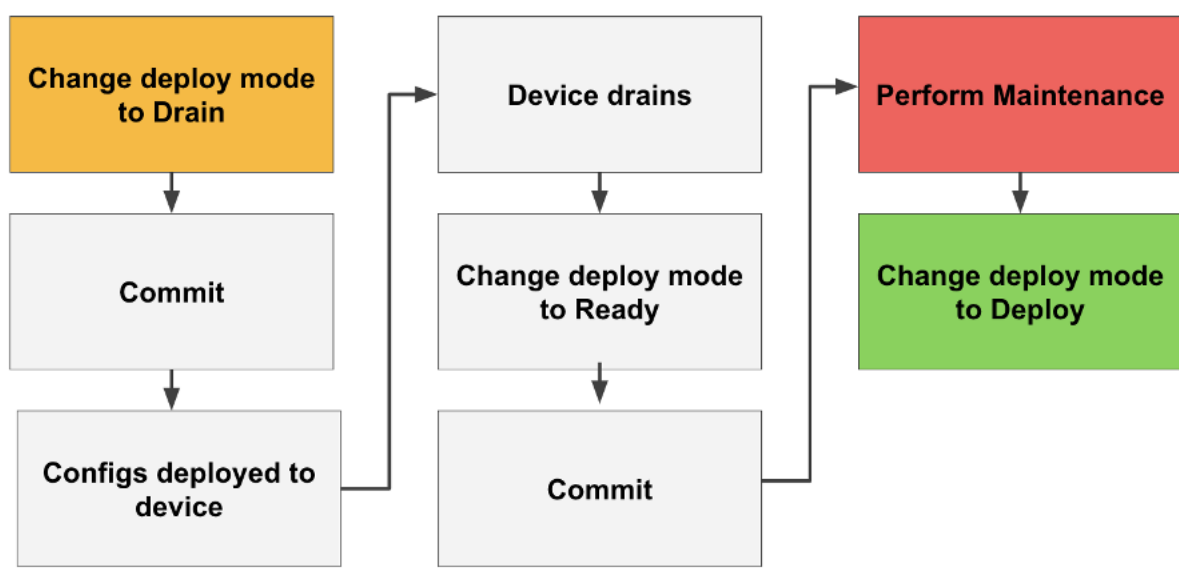
This article is derived from original documentation by Josh Saul

1.1 Introduction

This is implemented through modifications to the BGP process (inbound/outbound route-maps) and shutting down connected L2 server ports and MLAG peer link ports. By using Drain Mode, operators can minimize the number of dropped/lost traffic during these operations. During maintenance, redundancy is handled by ECMP/MLAG as long as there are suitable redundant systems in place. A visual example of Drain Mode on Spine switches is displayed below:

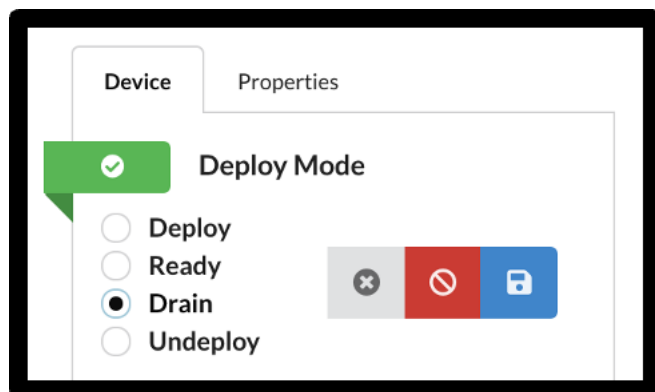


The workflow described in this document is summarized below:



1.2 Activating Drain Mode

Drain Mode is activated by switching devices to the Drain state in Apstra:



Once the device is switched to Drain, the change must be completed with the Commit button.

1.3 Disabling Drain Mode

To restore a device to service, switch the Deploy Mode setting back to Deploy, then commit.

1.4 IBA Monitoring of Devices in Drain Mode

You can activate a predefined IBA (Intent-Based Analytics) probe in Apstra, named “Drain Traffic Anomaly.” The required value for **Threshold in bps** works as follows:

- Value is the net sum of traffic on all hosted_interfaces
- This does not include traffic on the Ethernet management port which is not part of the probe measurement
- These interfaces include all L3 BGP-enabled paths
- Server-facing interfaces are shut during Drain Mode and are not part of this calculation
- The threshold describes the amount of traffic you wish to be alerted on (above the value) if devices are in the Drain state
- This ensures that you do not perform actual maintenance operations on a device that has not been fully drained.

1.5 Example

Spine1 is connected to 4 leaf switches, each connection runs the eBGP routing process. All application (server) based traffic flows will be rehashed via ECMP onto other links, the basic BGP neighbor updates will still be running. In a lab example with a small topology, this is effectively 1.5KBPS per link. With 4 neighbors, the total traffic we expect to remain on the devices is approximately 6KBPS. If we set the probe** Threshold in bps** to 10KBPS (10000), the probe will generate anomalies if there is more than 10K on all of the 4 interfaces combined.

1.5.1 Recommended Usage

Enable the probe with 100KBPS and leave it running in all blueprints. When a device enters the Drain state, you will see an anomaly as the traffic is removed from the links. This anomaly should only exist for a few seconds. If the anomaly does not clear, the device is not fully in Drain Mode. Once the anomaly clears, you can switch the device to the Ready state to take it out of service completely. You may not see the anomaly as it may appear and quickly disappear.

1.6 Configuration Examples

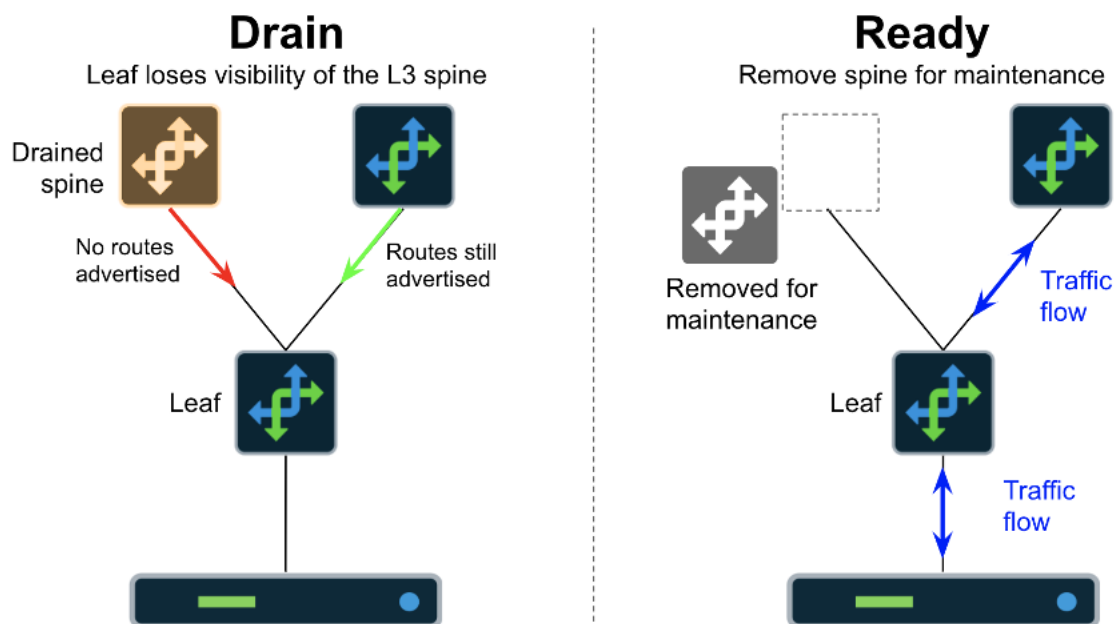
1.6.1 Spine (L2 and L3 Blueprints)

For this operation:

- A route-map is placed on the Spine switch on all BGP neighbors, restricting inbound and outbound routes

What happens:

- Outbound routes are removed from the device's routing table
- Routes to destinations with the device's ASN in the AS_PATH are removed from all devices in the network
- Packets are forwarded through the remaining ECMP paths for all destinations
- It is highly unlikely that a single in-flight packet will be lost. However, this is dependent on the L3 ECMP to L2 path hashing algorithms in the hardware and NOS



Drain NX-OS

```

1 ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
2 route-map Drain deny 10
3   match ip address prefix-list Drain
4   exit
5 !
6 neighbor 172.16.0.1 remote-as 64514
7   address-family ipv4 unicast
8     route-map Drain out

```

```

9      route-map Drain in
10      exit
11      exit
12      neighbor 172.16.0.3 remote-as 64514
13      address-family ipv4 unicast
14      route-map Drain out
15      route-map Drain in
16      exit
17      exit

```

Drain Junos

```

1  [edit policy-options]
2  + route-filter-list Drain {
3  + 0.0.0.0/0 upto /32;
4  + }
5  [edit policy-options]
6  + policy-statement Drain {
7  + term Drain-10 {
8  + from {
9  + family inet;
10 + route-filter-list Drain;
11 + }
12 + then reject;
13 + }
14 + }
15 [edit protocols bgp group l3clos-s neighbor 172.16.0.7]
16 + import ( Drain );
17 - export ( SPINE_TO_LEAF_FABRIC_OUT && BGP-AOS-Policy );
18 + export ( Drain );
19 [edit protocols bgp group l3clos-s neighbor 172.16.0.9]
20 + import ( Drain );
21 - export ( SPINE_TO_LEAF_FABRIC_OUT && BGP-AOS-Policy );
22 + export ( Drain );
23 [edit protocols bgp group l3clos-s neighbor 172.16.0.11]
24 + import ( Drain );
25 - export ( SPINE_TO_LEAF_FABRIC_OUT && BGP-AOS-Policy );
26 + export ( Drain );
27 [edit protocols bgp group l3clos-s-evpn neighbor 10.0.0.0]
28 + import ( Drain );
29 - export ( SPINE_TO_LEAF_EVPN_OUT );
30 + export ( Drain );
31 [edit protocols bgp group l3clos-s-evpn neighbor 10.0.0.1]
32 + import ( Drain );
33 - export ( SPINE_TO_LEAF_EVPN_OUT );
34 + export ( Drain );
35 [edit protocols bgp group l3clos-s-evpn neighbor 10.0.0.2]
36 + import ( Drain );
37 - export ( SPINE_TO_LEAF_EVPN_OUT );
38 + export ( Drain );

```

1.6.2 Leaf (L2 Server Facing Ports with MLAG)

For this operation:

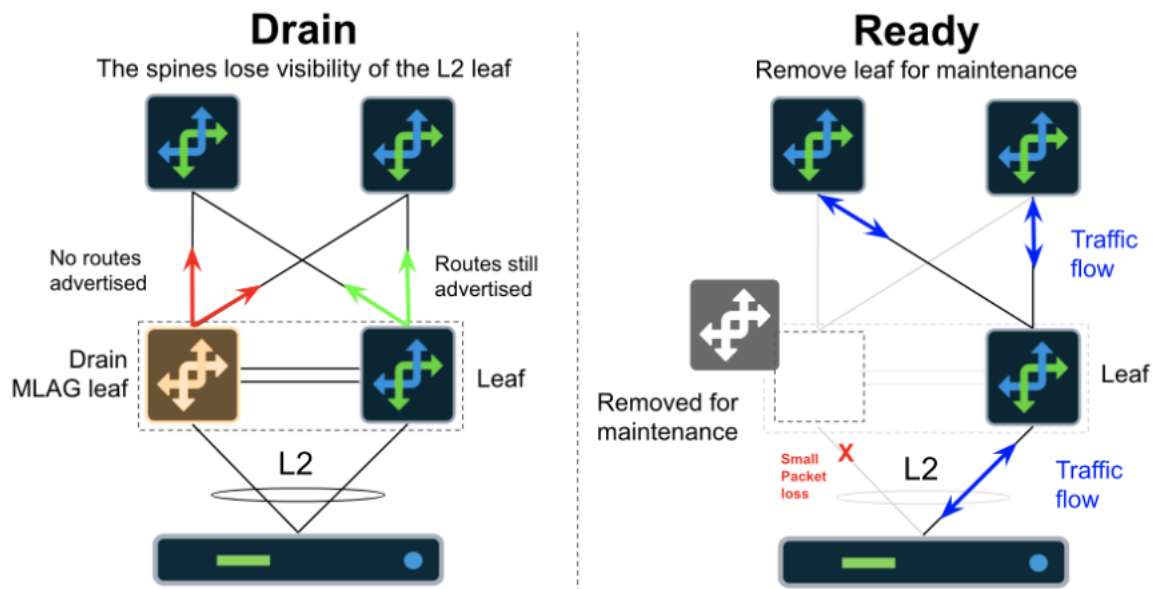
- A route-map is placed on all BGP neighbors restricting inbound and outbound routes
- Server facing interfaces are shutdown
- MLAG peer interfaces are shutdown

What happens (L3):

- Outbound routes are removed from the device's routing table
- Routes to destinations with the device's ASN in the AS-PATH are removed from all devices in the network
- Packets are forwarded through the remaining ECMP paths for all destinations
- It is highly unlikely that a single in-flight packet will be lost. However, this is dependent on the L3 ECMP to L2 path hashing algorithms in the hardware and NOS

What happens (L2):

- The server interface to this device will go DOWN
- Packets from the server that happen to be hashed onto this device via MLAG may be dropped depending on where they are in the forwarding process
- Packets from the server that happen to be hashed onto this device via MLAG may be forwarded over the MLAG peer link depending on where they are in the forwarding process
- Flows will be re-established on the alternate MLAG interfaces
- New flows will be established on the remaining MLAG interfaces



Drain (NX-OS)

```

1 interface Ethernet1/1
2   shutdown
3   exit
4 !
5 interface Ethernet1/2
6   shutdown
7   exit
8 !
9 interface port-channel1
10  shutdown
11  exit
12 !
13 ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
14 ipv6 prefix-list DrainV6 seq 5 permit 0::0/0 le 128
15 route-map Drain deny 10
16   match ip address prefix-list Drain
17   exit
18 !
19 route-map DrainV6 deny 10
20   match ipv6 address prefix-list DrainV6
21   exit
22 !
23 router bgp 64514
24   neighbor 10.0.0.0 remote-as 64512
25   address-family l2vpn evpn
26     route-map Drain out
27     route-map Drain in
28   exit
29   exit
30   neighbor 172.16.0.0 remote-as 64512
31   address-family ipv4 unicast
32     route-map Drain out
33     route-map Drain in
34   exit
35   exit

```

Drain (EOS)

```

1 interface Ethernet5
2   shutdown
3   exit
4   !
5 interface Ethernet6
6   shutdown
7   exit
8   !
9 interface port-channel1
10  shutdown
11  exit
12  !
13 interface port-channel2
14  shutdown
15  exit
16  !
17 ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
18 route-map Drain deny 10
19   match ip address prefix-list Drain
20   exit
21   !
22 router bgp 102
23   neighbor 10.10.4.0 route-map Drain out
24   neighbor 10.10.4.0 route-map Drain in
25   neighbor 10.10.4.8 route-map Drain out
26   neighbor 10.10.4.8 route-map Drain in
27   default neighbor 10.10.4.19 route-map MlagPeer out
28   neighbor 10.10.4.19 route-map Drain out
29   neighbor 10.10.4.19 route-map Drain in
30   !

```

Undrain (NX-OS)

What happens (L2):

- The server interface to this device will go UP
- New flows will be hashed onto the newly available MLAG interface

```

1 interface Ethernet1/1
2   no shutdown
3   exit
4   !
5 interface Ethernet1/2
6   no shutdown
7   exit
8   !
9 interface port-channel1
10  no shutdown
11  exit
12  !
13 no ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
14 no ipv6 prefix-list DrainV6 seq 5 permit 0::0/0 le 128
15 no route-map Drain deny 10
16   !
17 no route-map DrainV6 deny 10
18   !
19 router bgp 64514
20   neighbor 10.0.0.0 remote-as 64512
21   address-family l2vpn evpn
22     default route-map Drain out
23     default route-map Drain in
24   exit
25 exit

```

Undrain (EOS)

What happens (L2):

- Server interface to this device will go UP
- New flows will be hashed onto the newly available MLAG interface

```

1 interface Ethernet5
2   no shutdown

```

```

3  exit
4  !
5  interface Ethernet6
6    no shutdown
7  exit
8  !
9  interface port-channel1
10   no shutdown
11  exit
12  !
13  interface port-channel2
14   no shutdown
15  exit
16  !
17  no ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
18  no route-map Drain deny 10
19  !
20  router bgp 102
21   default neighbor 10.10.4.0 route-map Drain out
22   default neighbor 10.10.4.0 route-map Drain in
23   default neighbor 10.10.4.8 route-map Drain out
24   default neighbor 10.10.4.8 route-map Drain in
25   default neighbor 10.10.4.19 route-map Drain out
26   neighbor 10.10.4.19 route-map MlagPeer out
27   default neighbor 10.10.4.19 route-map Drain in
28  !

```

1.6.3 Leaf (L2 Server Facing Ports no MLAG)

For this operation:

- A route-map is placed on all BGP neighbors, restricting inbound and outbound routes
- Server-facing interfaces are shutdown

Drain (Junos)

```

1  [interfaces replace: ae1]
2  + disable;
3
4  [interfaces replace: xe-0/0/2]
5  + disable;
6
7  [interfaces replace: xe-0/0/3]
8  + disable;
9
10 [routing-instances blue protocols bgp group l3rtr neighbor 192.168.0.11]
11 - import ( RoutesFromExt-blue-Default_immutable );
12 - export ( RoutesToExt-blue-Default_immutable );
13 + import ( Drain );
14 + export ( Drain );
15
16 [routing-instances red protocols bgp group l3rtr neighbor 192.168.0.7]
17 - import ( RoutesFromExt-red-Default_immutable );
18 - export ( RoutesToExt-red-Default_immutable );
19 + import ( Drain );
20 + export ( Drain );
21
22 [protocols bgp group l3clos-l neighbor 172.16.0.2]
23 - export ( LEAF_TO_SPINE_FABRIC_OUT && BGP-AOS-Policy );
24 + import ( Drain );
25 + export ( Drain );
26
27 [protocols bgp group l3clos-l neighbor 172.16.0.8]
28 - export ( LEAF_TO_SPINE_FABRIC_OUT && BGP-AOS-Policy );
29 + import ( Drain );
30 + export ( Drain );
31
32 [protocols bgp group l3clos-l-evpn neighbor 10.0.0.3]
33 - export ( LEAF_TO_SPINE_EVPN_OUT && EVPN_EXPORT );
34 + import ( Drain );
35 + export ( Drain && EVPN_EXPORT );
36
37 [protocols bgp group l3clos-l-evpn neighbor 10.0.0.4]
38 - export ( LEAF_TO_SPINE_EVPN_OUT && EVPN_EXPORT );

```

```

39 + import ( Drain );
40 + export ( Drain && EVPN_EXPORT );
41
42 [protocols bgp group l3rtr neighbor 192.168.0.3]
43 - import ( RoutesFromExt-default-Default_immutable );
44 - export ( RoutesToExt-default-Default_immutable );
45 + import ( Drain );
46 + export ( Drain );
47
48 + [policy-options route-filter-list Drain]
49 + 0.0.0.0/0 upto /32;
50
51 + [policy-options policy-statement Drain term Drain-10 from]
52 + route-filter-list Drain;
53 + family inet;
54
55 + [policy-options policy-statement Drain term Drain-10]
56 + then reject

```

Drain (NX-OS)

```

1 interface Ethernet1/41
2   shutdown
3   exit
4   !
5 ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
6 route-map Drain deny 10
7   match ip address prefix-list Drain
8   exit
9   !
10 router bgp 64516
11   neighbor 172.16.0.8 remote-as 64512
12   address-family ipv4 unicast
13     route-map Drain out
14     route-map Drain in
15   exit
16   exit
17   neighbor 172.16.0.22 remote-as 64513
18   address-family ipv4 unicast
19     route-map Drain out
20     route-map Drain in
21   exit
22   exit
23   exit
24   !

```

Drain (EOS)

```

1 interface Ethernet5
2   shutdown
3   exit
4   !
5 ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
6 route-map Drain deny 10
7   match ip address prefix-list Drain
8   exit
9   !
10 router bgp 104
11   default neighbor 9.0.0.1 route-map RoutesToExt out
12   neighbor 9.0.0.1 route-map Drain out
13   default neighbor 9.0.0.1 route-map RoutesFromExt in
14   neighbor 9.0.0.1 route-map Drain in
15   neighbor 10.10.4.4 route-map Drain out
16   neighbor 10.10.4.4 route-map Drain in
17   neighbor 10.20.30.4 route-map Drain out
18   neighbor 10.20.30.4 route-map Drain in
19   neighbor 10.10.4.12 route-map Drain out
20   neighbor 10.10.4.12 route-map Drain in
21   neighbor 10.20.30.5 route-map Drain out
22   neighbor 10.20.30.5 route-map Drain in
23 vrf Finance
24   default neighbor 9.0.0.1 route-map RoutesToExt-Finance out
25   neighbor 9.0.0.1 route-map Drain out
26   default neighbor 9.0.0.1 route-map RoutesFromExt-Finance in
27   neighbor 9.0.0.1 route-map Drain in
28   exit
29   !

```

Undrain (NX-OS)

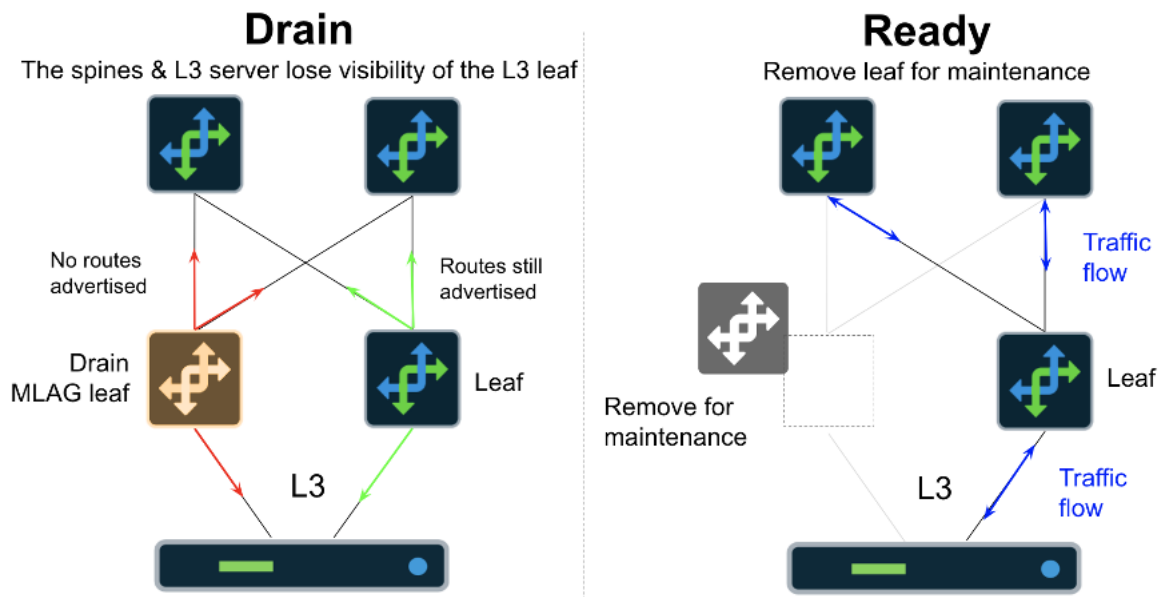
```
1 interface Ethernet1/41
2   no shutdown
3   exit
4   !
5   no ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
6   no route-map Drain deny 10
7   !
8   router bgp 64516
9     neighbor 172.16.0.8 remote-as 64512
10      address-family ipv4 unicast
11        default route-map Drain out
12        default route-map Drain in
13      exit
14    exit
15    neighbor 172.16.0.10 remote-as 64512
16      address-family ipv4 unicast
17        default route-map Drain out
18        default route-map Drain in
19      exit
20    exit
21    neighbor 10.0.0.1 remote-as 64513
22      address-family l2vpn evpn
23        default route-map Drain out
24        default route-map Drain in
25      exit
26    exit
27    neighbor 172.16.0.20 remote-as 64513
28      address-family ipv4 unicast
29        default route-map Drain out
30        default route-map Drain in
31      exit
32    exit
33    neighbor 172.16.0.22 remote-as 64513
34      address-family ipv4 unicast
35        default route-map Drain out
36        default route-map Drain in
37      exit
38    exit
39  exit
40  !
```

Undrain (EOS)

```
1 interface Ethernet5
2   no shutdown
3   exit
4   !
5   no ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
6   no route-map Drain deny 10
7   !
8   router bgp 104
9     default neighbor 9.0.0.1 route-map Drain out
10    neighbor 9.0.0.1 route-map RoutesToExt out
11    default neighbor 9.0.0.1 route-map Drain in
12    neighbor 9.0.0.1 route-map RoutesFromExt in
13    default neighbor 10.10.4.4 route-map Drain out
14    default neighbor 10.10.4.4 route-map Drain in
15    default neighbor 10.20.30.4 route-map Drain out
16    default neighbor 10.20.30.4 route-map Drain in
17    default neighbor 10.10.4.12 route-map Drain out
18    default neighbor 10.10.4.12 route-map Drain in
19    default neighbor 10.20.30.5 route-map Drain out
20    default neighbor 10.20.30.5 route-map Drain in
21    vrf Finance
22      default neighbor 9.0.0.1 route-map Drain out
23      neighbor 9.0.0.1 route-map RoutesToExt-Finance out
24      default neighbor 9.0.0.1 route-map Drain in
25      neighbor 9.0.0.1 route-map RoutesFromExt-Finance in
26    exit
27  !
```

1.6.4 Leaf (L3 Connected Servers)

For this operation: A route-map is placed on all BGP neighbors, restricting inbound and outbound routes



Drain (EOS)

```

1 ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
2 route-map Drain deny 10
3   match ip address prefix-list Drain
4   exit
5 !
6 router bgp 102
7   neighbor 10.10.4.0 route-map Drain out
8   neighbor 10.10.4.0 route-map Drain in
9   neighbor 10.10.4.8 route-map Drain out
10  neighbor 10.10.4.8 route-map Drain in
11  neighbor 11.0.0.1 route-map Drain out
12  neighbor 11.0.0.1 route-map Drain in
13 !
14 Undrain (EOS)
15 no ip prefix-list Drain seq 5 permit 0.0.0.0/0 le 32
16 no route-map Drain deny 10
17 !
18 router bgp 102
19   default neighbor 10.10.4.0 route-map Drain out
20   default neighbor 10.10.4.0 route-map Drain in
21   default neighbor 10.10.4.8 route-map Drain out
22   default neighbor 10.10.4.8 route-map Drain in
23   default neighbor 11.0.0.1 route-map Drain out
24   default neighbor 11.0.0.1 route-map Drain in
25 !

```

1.7 Summary

Generically, draining a device before taking it out of service is a standard operational procedure. Done “by hand,” it involves reconfiguring the device’s BGP so that traffic is directed away from the device – that is, BGP within a fabric sees the device as the least desirable next-hop to any destination. The practice ensures that packets in-flight through the device are delivered, while no new packets are sent through the device, minimizing packet drops before taking the device out of service.

Juniper Apstra quickly and easily automates the drain process, triggering the necessary configurations with a simple button click to put the device in Drain Mode and letting you know through a predefined IBA probe

when drain is completed, another click to undeploy the device, and another click to redeploy the device when maintenance is completed.

1.8 Useful links

- Apstra Drain Mode is also covered in the User Guide: <https://www.juniper.net/documentation/us/en/software/apstra4.1/apstra-user-guide/topics/task/device-drain.html>

1.9 Glossary

- AS_PATH: BGP attribute tracking Autonomous System Numbers along a specific route
- ASN: Autonomous System Number
- BGP: Border Gateway Protocol
- eBGP: External Border Gateway Protocol
- ECMP: Equal-Cost Multipath
- IBA: Intent-Based Analytics
- MLAG: (aka MC-LAG) Multi-Chassis Link Aggregation Group

1.10 Acknowledgements

This article is derived from original documentation by Josh Saul



DC – AI – Routing – Switching – Security – Wireless – Automation

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